

Exercise 1 SQL Fundamentals: Select and Filtering

1. SELECT *
FROM employees

Output

id	first_name	last_name	department	salary	hire_date	City
1	Alice	Green	IT	70 000	2020-10-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoeng	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

2. SELECT Distinct department
FROM employees ;

Output

department
IT
HR
Finance
Marketing

3. SELECT first_name,
last_name
FROM employees
ORDER BY salary DESC ;

Output

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Makana
Brain	Lee

```
4. SELECT id,  
           first_name,  
           last_name,  
           salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

Output

id	first_name	last_name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

```
5. SELECT id,  
           first_name,  
           last_name,  
           department  
FROM employees
```


~~WHERE employees = IT~~
WHERE department = 'IT';

Output

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. SELECT id,
first_name,
last_name,
department,
salary

FROM employees
WHERE department = 'Finance' AND salary > 60 000;

Output

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65 000

7. SELECT id,
first_name,
last_name,
department

FROM employees
WHERE department = 'HR' OR department = 'Marketing';

Output

id	first-name	last-name	department
1	Alice	Chen	HR
2	Brian	Lee	HR
4	David	Mokoena	Marketing

```
8. SELECT id,
        first-name,
        last-name,
        department
FROM employees
WHERE department NOT IN ('IT')
```

Output

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

```
9. SELECT id,
        first-name,
        last-name,
        department
FROM employees
WHERE department NOT IN ('IT')
WHERE department IN ('IT', 'HR', 'Finance')
```

id	first-name	last-name	department
----	------------	-----------	------------

1	Alice	Green	IT	70000
2	Brian	Lee	HR	65000
3	Cathy	Zulu	Finance	
5	Eva	Naidoo	IT	

10. SELECT id,
first_name,
last_name,
department,
salary,
city

FROM employees

WHERE ~~employees~~ department = 'IT' AND salary > 65000
AND city = 'Johannesburg'

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg